

Control work 2, and the quarter review

The whole derivative block in one paper, then the map that connects a sign chart to a sketch to an optimum.

LESSON 26–27

2 × 40 MIN

ALGEBRA 11, NAZORAT ISHI 2

P1 · CHAPTER 7 REVIEW

LESSON CLOCK

3

42'

10'

20'

12'

3

Instructions	3 min
The paper	42 min
Answers on the board	10 min
Rewrite	20 min
Concept map	12 min
Targets	3 min

LEARNING OBJECTIVES

- Apply the whole quarter in one assessment.
- Choose the correct rule and the correct test without prompting.
- Build a concept map of the derivative and its applications.
- Set a personal target for Quarter II.

TEXTBOOK

National	pp. 127–130
Cambridge	Pure Mathematics 1, pp. 191–194
Workbook	Control paper B

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Differentiate $y = x^4 - \frac{3}{x^2} + 2\sqrt{x}$	3	L7–9
2	Differentiate $y = x^2(3x - 1)^4$ and factorise	4	L10–12
3	Solve $ 2x - 5 \leq 7$ and $ x - 1 = 3x + 2 $	4	L15–16
4	Tangent and normal to $y = x^3 - 2x^2$ at $x = 2$	5	L17–18
5	Investigate $y = x^3 - 3x^2 - 9x + 2$ and sketch it	7	L19–22
6	A 36×36 card, corner squares cut: maximise the volume, with justification	7	L23–25

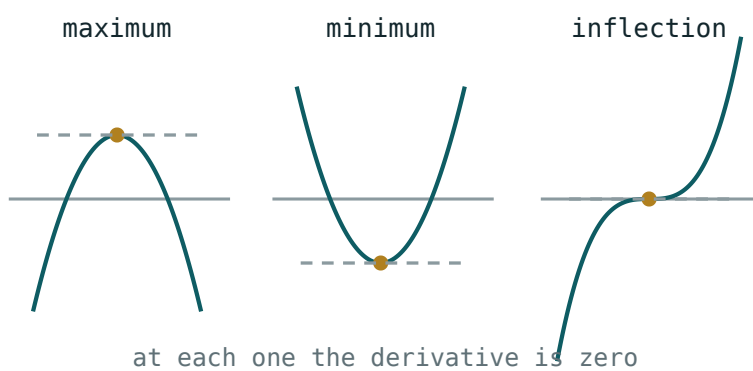
Q5 AND Q6 CARRY THE SIGN CHART

Q5 gives 3 marks for the stationary points and 4 for the chart, the nature and the sketch. Q6 gives 2 of its 7 for the justification alone. An unjustified optimum is an incomplete answer.

The concept map

Six boxes, links written as sentences:

- **increment** → **limit** — “shrink Δx and the quotient settles”
- **limit** → **derivative** — “the derivative is that limit”
- **derivative** → **rules** — “power, sum, product, quotient, chain”
- **derivative** → **tangent and normal** — “gradient at a point”
- **sign of f'** → **shape** — “positive climbs, negative falls, zero is stationary”
- **shape** → **optimisation** — “the extremum is where the gradient turns”



Three stationary points — the picture that links the sign chart to the sketch.

Looking forward

Quarter II opens with approximation by the tangent and with the indefinite integral — differentiation run backwards. Every rule of this quarter is used in reverse, so a rule half learned now costs twice as much then.

ONE HABIT TO CARRY FORWARD

Factorise f' before doing anything with it. Every sign chart, every stationary point, every optimisation in Quarter II starts from the factorised derivative.

02 Worked examples

EXAMPLE 1 Model answer, Q5: investigate $y = x^3 - 3x^2 - 9x + 2$.

① $y' = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$

② Stationary at $x = -1, 3$.

$$y(-1) = 7, y(3) = -25$$

③ $y'' = 6x - 6$

$$y''(-1) = -12 \text{ max}; y''(3) = 12 \text{ min}$$

④ Increasing outside $[-1, 3]$, decreasing inside.

ANSWER

max $(-1, 7)$, min $(3, -25)$

EXAMPLE 2 Model answer, Q6: 36×36 card, corner squares of side x .

① $V = x(36 - 2x)^2$

$$0 < x < 18$$

② $V' = (36 - 2x)(36 - 6x)$

③ $x = 6$ (or 18 , rejected).

④ $V' > 0$ before 6 , $V' < 0$ after.

Maximum.

ANSWER

$$V = 6 \times 24^2 = 3456 \text{ cm}^3$$

EXAMPLE 3 Model answer, Q2: differentiate $y = x^2(3x - 1)^4$.

① Product, with the chain rule inside.

② $y' = 2x(3x - 1)^4 + x^2 \cdot 4(3x - 1)^3 \cdot 3$

③ $= 2x(3x - 1)^3[(3x - 1) + 6x]$

④ $= 2x(3x - 1)^3(9x - 1)$

ANSWER

$$2x(3x - 1)^3(9x - 1)$$

03 Interactive model

Work Q5 on the board with the sign chart drawn under the sketch.

The quarter in ten questions

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} :$$

0

2

4

undefined

First principles, $(x^3)'$:

 $3x$ $3x^2$ x^2 $3x^3$

$(\frac{1}{x^2})'$:

 $-\frac{2}{x^3}$ $\frac{2}{x^3}$ $-\frac{1}{x^3}$ $-2x$

$((3x - 1)^4)'$:

Solve $|2x - 5| \leq 7$:

Tangent to $y = x^3 - 2x^2$ at $x = 2$ has gradient:

Stationary points of $y = x^3 - 3x^2 - 9x$:

$f' = 0, f'' > 0$ means:

maximum

minimum

inflection

nothing

36×36 card: optimal corner cut is:

4

6

9

12

On a closed interval the maximum can be at:

a stationary point only

an endpoint only

either

neither

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Command word	Topshiriq so'zi	Командное слово
Justification	Asoslash	Обоснование
Self-assessment	O'z-o'zini baholash	Самооценка
Target	Maqsad	Цель
Revision	Takrorlash	Повторение
Summative assessment	Yakuniy baholash	Итоговое оценивание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Show that” requires:

the answer

every line

a sketch

a calculator

An optimisation answer without justification loses:

nothing

a mark or two

all marks

half

Before using a sign chart you should:

sketch

factorise f'

find f''

substitute

Quarter II begins with:

trigonometry

the integral

probability

vectors

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(x^4)'$

ANSWER $4x^3$

2. $(\frac{3}{x^2})'$

ANSWER $-\frac{6}{x^3}$

3. $(2\sqrt{x})'$

ANSWER $\frac{1}{\sqrt{x}}$

4. Solve $|2x - 5| \leq 7$

ANSWER $-1 \leq x \leq 6$

5. Gradient of $y = x^3 - 2x^2$ at $x = 2$

ANSWER 4

6. Stationary points of $y = x^2 - 4x$

ANSWER $x = 2$

7. 36×36 card: best corner cut

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. $(x^4 - \frac{3}{x^2} + 2\sqrt{x})'$

ANSWER $4x^3 + \frac{6}{x^3} + \frac{1}{\sqrt{x}}$

2. $(x^2(3x - 1)^4)'$

ANSWER $2x(3x - 1)^3(9x - 1)$

3. Solve $|x - 1| = |3x + 2|$

ANSWER $x = -0.25$ or $x = -1.5$

4. Tangent to $y = x^3 - 2x^2$ at $x = 2$

ANSWER $y = 4x - 8$

5. Normal there

ANSWER $y = -0.25x + 0.5$

6. Investigate $y = x^3 - 3x^2 - 9x + 2$

ANSWER max $(-1, 7)$, min $(3, -25)$

7. 36×36 card: greatest volume

ANSWER 3456 cm^3

Hard · stretch and reason

7 PROBLEMS

1. $\left(\frac{x^2}{\sqrt{x+1}}\right)'$

ANSWER $\frac{x(3x+4)}{2(x+1)^{3/2}}$

2. Find a, b so $y = x^3 + ax^2 + bx$ has a max at $x = -1$ and min at $x = 2$

ANSWER $a = -1.5, b = -6$

3. Solve $|x - 2| + |x + 3| = 7$

ANSWER $x = -4$ or $x = 3$

4. Tangent to $y = x^3$ at $x = a$ meets the curve again where?

ANSWER $x = -2a$

5. Closed cylinder volume 2000 cm^3 : minimise surface area

ANSWER $r \approx 6.83 \text{ cm}$

6. Greatest value of $y = x^2(6 - x)$ on $[0, 6]$

ANSWER 32 at $x = 4$

7. Show that between two roots of a differentiable function there is a stationary point

ANSWER Rolle's theorem — the curve must turn

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter II.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. Investigate $y = x^3 - 6x^2 + 9x + 1$ fully and sketch it.
4. Write your target for Quarter II in one checkable sentence, and date it.